

Alexander Nardi

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EDUCATION

University of Central Florida, Burnett Honors College

Bachelor of Science in Computer Science

Orlando, FL

Expected Graduation: May 2027

EXPERIENCE

AI/ML Engineering Intern

June 2026 – Sept. 2026

FTI Defense - Frontier Technology Inc.

Remote, USA

- Conceived and built a drop-in **Python obfuscation framework** for Dockerized applications that resists LLM-assisted reverse engineering while preserving original behavior, with near-zero runtime overhead and a **52%** reduction in storage footprint.
- Validated resistance to LLM-assisted reverse engineering through adversarial white-box testing, measuring **~380×** greater attack time and **~275×** greater token usage, with only **~35%** of functionality reconstructed despite exposed public APIs.
- Productionized the core **~3,600-line** obfuscation framework into a shared internal library, resulting in adoption and deployment across live production software.
- Hand-selected by the CTO to provide technical feedback and recommend fixes ahead of a client-facing demo, identifying and documenting **150+** reproducible issues across UI/UX, functionality, and API integrations.

Programming Lead

Aug. 2025 – Present

VEX U Robotics Competition: UCF7 Daydream

Orlando, FL

- Led **10+** programmers by defining semester-wide technical vision, structuring development into coordinated workstreams, and directing software architecture and hardware integration to deliver **2** competition robots.
- Led **7** technical workshops and structured onboarding for new programmers, retaining **5 of 7** as active project contributors.
- Designed a task-specific **heatmap CNN** for CPU-only **robot vision**, deriving a custom focal-weighted classification loss and offset-regression head tailored for precise sub-grid object localization.
- Automated the generation and labeling of **60,000-image** CNN training sets using an **NVIDIA NDDS** synthetic data pipeline.
- Deployed the model on a **Raspberry Pi 5**, achieving **99.7% precision**, **96.8% recall**, and **~50 ms CPU inference latency**.
- Reduced path-following error by **27%** using a custom **C++** cubic-spline interpolation framework, enabling fine-motion control.

Embedded Software Engineer

Jan. 2025 – Aug. 2025

VEX U Robotics Competition: UCF7 Daydream

Orlando, FL

- Engineered **C++** control software for two robots, integrating sensor fusion and operator control systems that helped drive UCF7 to a **Top-20** U.S. ranking and a qualification for the 2024-2025 **VEX U World Championship**.
- Collaborated with mechanical and electrical subteams on sensor-placement and motor-control refinements, improving autonomous motion consistency by **14%**.
- Developed quaternion-based PID controllers with $\pm 0.1^\circ$ orientation precision, improving trajectory reliability by over **30%**.

PROJECTS

DAVE | 1st Place Overall | *Python, scikit-learn, Gemini API, FastAPI, React, Embedded Systems*

Jul. 2026

- Built a wearable motion-tracking volleyball coach with a **4-person** team in 12 hours, earning **1st place** at BloomKnights among **86 projects and 248 participants**.
- Automated swing analysis with an end-to-end ML pipeline processing synchronized dual-IMU telemetry at **200 Hz**, reconstructing arm motion through a two-link kinematic model, and extracting **33** biomechanical features for **Random Forest** classification.
- Delivered real-time athlete coaching through a **Gemini API** system generating swing-specific feedback from ML classifications and 3D biomechanical metrics, paired with interactive **3D motion replay** for visual analysis.

Growcery | *PyTorch, FastAPI, Docker, Gemini 2.5 Flash*

Oct. 2025

- Enabled real-time agricultural monitoring alongside **3** teammates by integrating **CNN visual inference** with **Gemini 2.5 Flash** reasoning to assess produce freshness and crop health.
- Fine-tuned **EfficientNetV2** with transfer learning and data augmentation for crop and produce disease detection, achieving **98.9%** validation F1 score.
- Achieved **sub-15 ms** inference through a FastAPI backend on an **AMD ROCm GPU server**, feeding CNN predictions into Gemini ADK for real-time reasoning.

TECHNICAL SKILLS

Languages: Python, C/C++, TypeScript, JavaScript, Java, C#

AI/ML: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, Gemini API, Gemini ADK, ROCm, NumPy, Pandas

Embedded/Robotics: Raspberry Pi, ESP32, PlatformIO, Arduino, PROS, Sensor Fusion, PID Control, Odometry

Developer Tools: VS Code, Git, GitHub, GitHub Projects, Azure DevOps, Google Colab, Linux, WSL, Docker, FastAPI